

PoC Proposal for XBRL GL Next: A Hierarchical Tidy Data Universal Audit Data Converter for Invoice Reuse

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Subject	PoC proposal for a hierarchical tidy data Universal Audit Data Converter (UADC)
GitHub	https://github.com/pontsoleil/UADC-PoC/

Abstract

This paper proposes an early Proof of Concept for XBRL GL Next based on a Universal Audit Data Converter (UADC) using hierarchical tidy data as the common semantic representation. The proposed PoC converts OpenPeppol invoice XML, together with a syntax binding definition in CSV, into hierarchical tidy data CSV for the EN 16931 core invoice semantic model based on the XBRL GL Next OIM taxonomy. The semantic invoice model represents both document-level information and line-item information. Accordingly, the converted invoice has materialized as related invoice CSV and line CSV datasets, together with reusable seller and buyer instances across multiple invoices. From that common semantic dataset, semantic binding definitions in CSV are used to generate downstream outputs including an ISO 21378:2019 Audit data collection Sales/Purchase invoice view and AICPA ADS (Audit Data Standard) O2C/P2P invoice views. For the AICPA target, the PoC recognizes support for both pipe-delimited flat files and XBRL GL in XBRL 2.1 tuple syntax. The PoC is intended to demonstrate deterministic transformation, semantic reuse, traceability, and audit-oriented interoperability.

Background

The current XBRL GL Next direction emphasizes that transactional information remains fragmented across software platforms, data formats, organizational boundaries, and reporting domains, while users increasingly require reusable access to the underlying business events from which reports are derived.

The Transactional Reporting Working Group Charter calls for modernization of XBRL GL, interoperable transactional reporting, assurance-ready data, trusted data exchange, AI and automation readiness, and practical Proofs of Concept demonstrating traceability and interoperability across reporting domains.

The XBRL GL Next Requirements Specification further defines XBRL GL Next as the core semantic data file or dataset for a Universal Adapter pattern connecting EDI, ERP, accounting, audit, tax, statistical, and reporting systems. It also requires hierarchically meaningful, self-explanatory business data, deterministic transformation, continuity with existing XML-based implementations, and OIM-based realizations such as xBRL-CSV.

For these reasons, a UADC PoC is not an isolated experiment. It is a direct test of whether XBRL GL Next can provide the common semantic layer needed for invoice transformation, audit reuse, and future multi-domain interoperability. Because the semantic representation is based on the EN 16931 core invoice semantic model, the same approach can also be used to map proprietary domestic e-invoices to an EN 16931 CIUS.

Proposal

The proposed PoC scenario is the following formal conversion pattern:

- OpenPeppol invoice XML + Syntax binding definition in CSV
 - UADC
 - hierarchical tidy data CSV for the EN 16931 core invoice semantic model based on the XBRL GL Next OIM taxonomy
- Hierarchical tidy data CSV based on the XBRL GL Next OIM taxonomy + Semantic binding definition in CSV
 - UADC
 - ISO 21378:2019 Audit data collection Sales/Purchase invoice view
- Hierarchical tidy data CSV based on the XBRL GL Next OIM taxonomy + Semantic binding definition for piped ADS in CSV
 - UADC
 - AICPA ADS (Audit Data Standard) O2C/P2P invoice view, in pipe-delimited flat-file form
- Hierarchical tidy data CSV based on the XBRL GL Next OIM taxonomy + Semantic binding definition for XBRL GL tuple ADS in CSV
 - UADC
 - AICPA ADS (Audit Data Standard) O2C/P2P invoice view, in XBRL GL in XBRL 2.1 tuple form

In this architecture, syntax binding definitions support XML-to-CSV conversion from the source syntax into a common semantic representation. The result of syntax binding is hierarchical tidy data CSV for the EN 16931 core invoice semantic model, including buyer and seller data across multiple invoices.

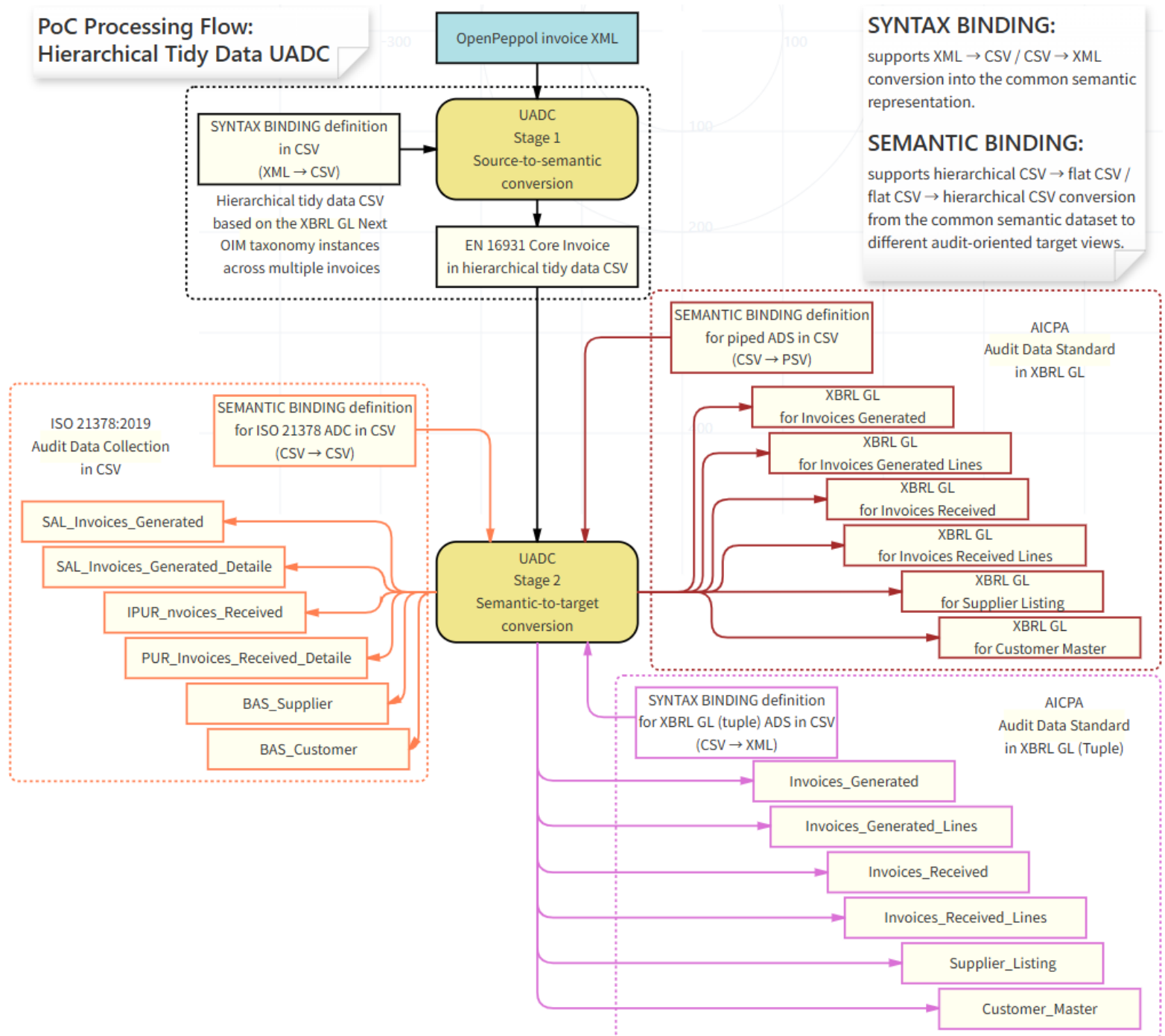
Semantic binding definitions in CSV then support CSV-to-CSV conversion from that common semantic representation into separated target CSVs and other audit-oriented target views. The same invoice data can therefore produce SAL_Invoices_Generated for the seller side and PUR_Invoices_Received for the buyer side. Where line-level separation is required, the same semantic source can also produce the corresponding detail outputs. For supplier-oriented audit data, buyer information in the invoice is transformed into customer-oriented target CSVs. For buyer-oriented audit data, seller information in the invoice is transformed into supplier-oriented target CSVs.

The key design principle is that no source syntax should become the internal canonical model. Instead, the source invoice is mapped into a common semantic representation, and all downstream outputs are generated from that representation.

Processing Flow

Figure 1 summarizes the proposed two-stage UADC process. Stage 1 transforms OpenPeppol invoice XML into the common semantic representation, where the EN 16931 core invoice semantic

model materialized as hierarchical tidy data CSV. That representation includes related invoice CSV and line CSV datasets, together with reusable sales invoice, purchase invoice, supplier, and customer instances across multiple invoices. Stage 2 reuses that semantic dataset, through semantic binding definitions in CSV, to produce downstream views including ISO 21378:2019 Audit data collection



output and AICPA ADS O2C/P2P output. For the AICPA side, the PoC should allow realization either as pipe-delimited flat files or as XBRL GL in XBRL 2.1 tuple syntax.

Figure 1. Processing flow for the proposed hierarchical tidy data UADC PoC, including EN 16931 core invoice semantic-model generation, separated invoice CSV and line CSV datasets, ISO 21378:2019 Audit data collection outputs, and alternative AICPA ADS output forms.

Functional Architecture

1. Source-side conversion

OpenPeppol invoice XML is combined with a syntax binding definition in CSV. The syntax binding definition identifies how physical source XML elements are mapped to the semantic items and hierarchical structures of the XBRL GL Next OIM model. It also supports generation of hierarchical tidy data CSV for the EN 16931 core invoice semantic model, including the decomposition of the source invoice into related invoice CSV and line CSV datasets.

2. Common semantic layer

The result is hierarchical tidy data CSV based on the XBRL GL Next OIM taxonomy. The semantic invoice model represents both invoice document information and invoice line items. Accordingly, the converted invoice consists of related invoice CSV and line CSV datasets. This intermediate dataset is not a flat extract. CIUS does not itself distinguish between sales invoices and purchase invoices. However, the same invoice can be interpreted as a sales invoice from the seller's perspective and as a purchase invoice from the buyer's perspective.

It preserves grouped business meaning and represents seller-side sales invoice instances, buyer-side purchase invoice instances, supplier instances, and customer instances across multiple invoices.

3. Target-side conversion

Semantic binding definitions in CSV are then applied to the common semantic dataset. Their role is not to reinterpret business meaning, but to reorganize and project the semantic dataset into target formats required by different audit frameworks. They support separate CSV outputs in different target formats, including document-level and line-level outputs where required. In particular, the same invoice data can be projected as SAL_Invoices_Generated for the seller view and as PUR_Invoices_Received for the buyer view.

4. Converter roles

Syntax binding supports XML-to-CSV conversion into the common semantic representation, including generation of separated invoice CSV and line CSV datasets.

Semantic binding supports CSV-to-CSV conversion from the common semantic dataset to separated or reformatted target CSVs and other target views, including ISO 21378:2019 invoice views and AICPA ADS O2C/P2P views. The same invoice data can therefore support both seller-side and buyer-side audit views from one common semantic source.

The UADC therefore separates physical syntax transformation from semantic re-expression. It also separates formation of the common semantic invoice representation from production of downstream target layouts and serializations.

Why Hierarchical Tidy Data

Ordinary tidy data is valuable because it supports machine processing, validation, and reuse. However, transactional business data is not only tabular. It is also hierarchical, contextual, repeated, and linked across multiple business objects.

A business invoice contains document-level information, parties, references, tax subtotals, and line items. Across multiple invoices, sellers and buyers recur as reusable semantic objects. In this PoC, the semantic invoice model represents both document and line items, while the converted invoice

materialized as related invoice CSV and line CSV datasets. If this information is reduced to disconnected flat tables or reassembled only through local code and joins, meaning must be reconstructed externally, which weakens traceability, reuse, and auditability.

Hierarchical tidy data extends the tidy data idea by preserving explicit hierarchy and semantic grouping while retaining machine processability. In this PoC, that means invoice structure remains explicit, repeated parties can be represented as reusable instances, invoice and line datasets can be generated without losing their semantic relationship, and downstream views can be generated deterministically.

PoC Scope and Instance Model

The PoC should explicitly support at least the following semantic instance groups across multiple invoices:

- sales/purchase invoice instances at document level
- supplier instances and customer instances
- invoice line instances and tax subtotals
- document references, identifiers, dates, currencies, and monetary amounts

This multi-invoice scope makes the PoC stronger than a single-document transformation. It demonstrates that the common semantic representation is a reusable transaction-level dataset rather than a one-off interchange snapshot. It also demonstrates that semantic binding definitions can transform the common representation into separated invoice CSV and line CSV structures without loss of meaning.

Deliverables

- sample OpenPeppol invoice XML instances
- a syntax binding definition in CSV for source XML mapping
- a hierarchical tidy data CSV dataset based on the XBRL GL Next OIM taxonomy, including related invoice CSV and line CSV datasets
- instances for sales/purchase invoices, suppliers, and customers across multiple invoices
- semantic binding definitions in CSV for target view generation, including separated CSV outputs where required
- an ISO 21378:2019 Audit data collection Sales/Purchase invoice output
- AICPA ADS (Audit Data Standard) O2C/P2P outputs in pipe-delimited flat-file form and/or XBRL GL in XBRL 2.1 tuple form
- a traceability demonstration from source invoice facts to semantic items and then to each target view
- a short lesson-learned note on semantic coverage, mapping gaps, reuse patterns, and extension needs

Next Steps

- Agree the constrained business scope and the minimum dataset for the first PoC iteration.
- Define the XBRL GL Next profile, including relevant modules, roots, identifiers, and validation constraints.

- Prepare representative OpenPeppol invoice XML samples and the initial syntax binding CSV.
- Define the initial hierarchical tidy data CSV model, including invoice CSV and line CSV datasets, and prepare the semantic binding CSV definitions for the target views.
- Implement Stage 1 and Stage 2 transformation logic in the UADC.
- Validate deterministic transformation, traceability, and reuse across multiple invoices.
- Document lessons learned for architecture, profile design, extension governance, separated CSV generation, and future standardization.

References

- Transactional Reporting Working Group Charter Statement (2026)
- XBRL GL Vision Statement (2026)
- XBRL GL Next Requirements Specification (2026)
- XBRL GL Next Taxonomy Framework — Part 1: General rules
- XBRL GL Next Taxonomy Framework — Part 3: xBRL-CSV palette taxonomy

